VCS VERIFICATION REPORT

QINGDAO LONGXIN WIND POWER PROJECT PHASE I

(VCS PROJECT ID: 969)

Document Prepared By

Shenzhen CTI International Certification Co., Ltd

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Summary:

Shenzhen CTI International Certification Co., Ltd (CTI) has performed the verification of the emission reductions reported for the project activity "Qingdao Longxin Wind Power Project Phase I" (VCS Project ID:969) for the monitoring period 21/12/2017-07/11/2021 within the 1st VCS crediting period 08/11/2011-07/11/2021, to review and determine the monitored reductions in GHG emissions that have occurred as a result of the project activity. These emission reductions are claimed as Verified Carbon Units (VCU) under the VCS Standard Version 4.3.

The verification was performed on the basis of VCS Standard Version 4.3 for the VCS projects, as well as criteria given to provide for consistent project operations, monitoring and reporting. The verification was conducted by means of document review, follow-up interviews and site inspections, and the resolution of outstanding issues. The verification team identified one CL, and no CAR or FAR is raised in this monitoring period.

In CTI's opinion, the GHG emission reductions reported for the project in the monitoring report (version 02 dated 08/09/2022) are fairly stated. The GHG emission reductions of the period from 21/12/2017-07/11/2021 within the 1st VCS crediting period were calculated correctly on the basis of approved methodology ACM0002 "Consolidated methodology for grid-connected electricity generation from renewable sources" (version 12.2.0), the registered VCS PD (version 02, dated

30/01/2013) and the monitoring plan contained in the registered CDM-PDD (version 03 dated 18/04/2012)

CTI does not assume any responsibility towards the issuance and utilization of the VCUs hereby verified and certified. Request for issuance of VCUs shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to CTI and the engagement conditions detailed in this report. CTI cannot be held liable by any party for decisions made or not made based on this report.

Hence, CTI is able to certify that the emission reductions from the "Qingdao Longxin Wind Power Project Phase I" during this monitoring period from 21/12/2017-07/11/2021 amount to 366,046 tCO_2e.

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1 INTRODUCTION

Juno Carbon Investment & Environmental Technology (Beijing) Co., Ltd. has commissioned Shenzhen CTI International Certification Co., Ltd (CTI) to carry out the verification and certification of emission reductions reported for the "Qingdao Longxin Wind Power Project Phase I" (the Project) for the monitoring period 21/12/2017-07/11/2021. This report contains the findings from the verification and includes a verification statement for the verified carbon units.

1.1 Objective

Verification is the periodic independent review and ex-post determination by an accredited verification body of the monitored reductions in GHG emissions that have occurred as a result of the registered VCS project activity during a defined verification period.

A verification statement is the written assurance by a verification body that, during a specific period in time, a project activity achieved the emission reductions as verified.

The objective of this verification was to verify and provide a verification statement of emission reductions reported for the "Qingdao Longxin Wind Power Project Phase I" for the period 21/12/2017-07/11/2021.

1.2 Scope and Criteria

The scope of the verification is:

- To verify that actual monitoring systems and procedures are in compliance with the monitoring systems and procedures described in the monitoring plan;
- To evaluate the GHG emission reduction data and express a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement;
- To verify that reported GHG emissions data is sufficiently supported by evidence.

The criteria of the verification are:

- VCS Program Guide (version 4.2)/33/
- VCS Standard (version 4.3)/32/ and other relevant requirements defined by Verra;
- The approved methodology ACM0002 (version 12.2.0)/36/ applied by the project.

The verification shall ensure that reported emission reductions are complete and accurate in order to be verified.

1.3 Level of Assurance

The verification report expresses a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement. As per the para 3.9.1 of the VCS Standard Version 4.3, the project size categorization belongs to 'Project' as the expected annual emission reductions is 92,183 tCO_2e lower than 300,000 tCO_2e. Therefore, according to the requirement of para 4.1.8 (4) of VCS standard Version 4.3 "The threshold for materiality with respect to the aggregate of errors, omissions and misrepresentations relative to the total reported GHG emission reductions and/or removals shall be five percent for projects and one percent for large projects.", CTI applied a materiality threshold of 5% with respect to omission or misstatements concerning reported quantities of the project activity.

1.4 Summary Description of the Project

Sectoral Scope and Project Type

According to the VCS Program Guide (version 4.2)/33/ and Annex A of the Kyoto Protocol, the project is applicable under the following activity categories:

- Sectoral Scope: 1. Energy (Renewable/non-renewable)
- Project Type: Grid-connected wind power project

Project Background

The project is located in Liyuan Town and Dianzi Town, Pingdu City, Shandong Province of P.R.China. The site location's approximate coordinates are 119.8827ºE to 119.9503ºE and 36.8157ºN to 37.8915ºN and the central geographical coordinates are 119.91125ºE and 36.8568ºN. The project activity was registered as a CDM project with Ref.6697 on 28/09/2012 and the renewable crediting period was selected starting on 28/09/2012. The project has been registered as VCS project with VCS ID. 969 under VCS Standard Version 3 with the crediting period start date 08/11/2011 when the commencement of operations activities started as confirmed by checking the Commissioning Report /26/, Daily operation and maintenance records/7/, the VCS monitoring report and verification report for the period of 08/11/2011-27/09/2012 /23//24/.

The amount of 82,302 tCO_2e VCUs have been issued for the monitoring period 08/11/2011-27/09/2012 within the 1st VCS crediting period. GHG emission reductions have not been applied for CERs issuance under CDM. The installed capacity of the project is 49.8MW equipped with 24 sets of wind turbine generator in type of V90-2.0MW, with unit capacity of 2MW and 1 set of wind turbine generator in type of V90-1.8MW, with unit capacity of 1.8MW, all manufactured by Vestas Wind Systems A/S The electricity generated by the project activity is supplied to the North China Power Grid (NCPG), and the project is estimated to deliver 92,183 tCO_2e emission reductions annually.

2 VERIFICATION PROCESS

2.1 Method and Criteria

The verification was performed through means of the following three phases in accordance with the requirement of the registered VCS PD (version 02 dated 30/01/2013) and monitoring plan contained in the registered CDM PDD, the applied methodology, and the VCS Standard (Version 4.3) and other relevant VCS requirements:

- A desk review of the monitoring report and all support documents;
- Follow-up interviews with project stakeholders and site inspection;
- The resolution of outstanding issues and the issuance of the verification report and statement.

The following sections outline each step in more detail.

The verification of the emission reductions has assessed all factors and issues that constitute the basis for emission reductions from the project. These include:

- The emission reduction calculations and the relevant data records;
- The calibration and maintenance records for the monitoring instruments;
- The management systems to support the project operation and monitoring.

2.2 Document Review

Based on the requirements of competency, experience and qualified sectoral scopes, CTI appointed a verification team in accordance with CTI's internal procedures.

Function	Name	Technical competence	Task Performance*
Team Leader& Verifier	Feng Tian	1.2	☒DR ☒SV ☒RP ☐TR
Technical Reviewer	Lin Shunrong	1.2, 14.1, 15.1	☒DR ☐SV ☐RP ☒TR

*DR=Document review; SV=Site visit; RP=Reporting; TR=Technical review

In addition to the VER/VCU monitoring report/1/, the registered VCS PD /11/, emission reduction calculation spreadsheet/2/ during this monitoring period, the following documents also were assessed as a part of the verification audit:

- The CDM registered PDD and corresponding validation report /22/;
- The previous VER/VCU monitoring report/23/ and verification report/24/
- Baseline and monitoring methodology ACM0002 (version 12.2.0) applied by the project/36/;

- Relevant decisions, clarifications and guidance from the Verra/32/-/35/; and
- Other information and references relevant to the project activity.

During the desk review, CTI has applied standard auditing techniques to assess the quality of information provided. The following activities were performed:

- A review of the data and information presented to verify their completeness;
- A review of the monitoring plan and monitoring methodology, paying particular attention to the frequency of measurements, the quality of metering equipment including calibration requirements, and the quality assurance and quality control procedures; and
- An evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions.

2.3 Interviews

On 09/08/2022, CTI visited the project proponent China Resources Wind Power (Qingdao) Co., Ltd. to perform on-site assessment. The key personnel of the project were interviewed or assisted the verification team /37/. Main topics of the interview cover implementation of the project construction, applicability of selected methodology, implementation of project monitoring, emission reduction calculation, etc.

The key personnel interviewed /37/ are summarized in the table below:

<table>
<tr><th>Interviewed personnel</th><th>Role</th><th>Organization</th><th>Subject</th></tr>
<tr><td>Mr. Qiu Zhenzhen</td><td>Plant Director</td><td rowspan="3">China Resources Wind Power (Qingdao) Co., Ltd.</td><td rowspan="3">Operation of the project activity;
Implementation of the monitor plan of the project activity;
Data collection and data achievement;
Calibration of meters and equipment maintenance;
The complaint by local stakeholders;
The stakeholder consultation during the operation of the project activity.
Project's sustainable development contributions.</td></tr>
<tr><td>Mr. Li Jiazhuo</td><td>O&M Staff</td></tr>
<tr><td>Mr. Zhang Shufeng</td><td>O&M Staff</td></tr>
<tr><td>Mr. Man Wenyan</td><td rowspan="4">Local resident</td><td rowspan="4">Local Village</td><td rowspan="5">The impact of the project activity;
The complaint by local stakeholders;
The stakeholder consultation during the operation of the project activity;
Project's sustainable development contributions.</td></tr>
<tr><td>Ms. Tanliyun</td></tr>
<tr><td>Mr. Yang Shaokun</td></tr>
<tr><td>Mr. Hou Kun</td></tr>
<tr><td>Mr. Li Peiyu</td><td>Staff</td><td>Local Environment Protection Bureau</td></tr>
<tr><td>Ms. Ding Xue</td><td>Project Manager</td><td>Juno Carbon Investment & Environmental Technology (Beijing) Co., Ltd.</td><td>Data collection and ER calculation.</td></tr>
</table>

2.4 Site Inspections

During the on-site assessment, CTI has applied standard auditing techniques to assess the quality of information provided. The following aspects of the project activity have been verified:

- An assessment of the implementation and operation of the registered project activity is as per the registered VCS PD of the project activity;
- A review of information flows for generating, recording, aggregating and reporting the monitoring parameters; and
- Interviews with relevant personnel to determine whether the operational and data collection procedures are implemented in accordance with the monitoring plan;
- A cross-check between information provided in the monitoring report and data from other sources such as plant logbooks and the electricity receipts;
- A check of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the registered VCS PD and the selected methodology;
- A review of calculations and assumptions made in determining the GHG data and emission reductions; and
- An identification that quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

In addition, all parameters required by the monitoring methodology ACM0002 (version 12.2.0), and the management system were assessed during the site visit.

2.5 Resolution of Findings

A corrective action request (CAR)shall be raised, where:

i. Non-conformities with the monitoring plan or methodology are found in monitoring and reporting, or if the evidence provided to prove conformity is insufficient;

ii. Mistakes have been made in applying assumptions, data or calculations of emission reductions which will impair the estimate of emission reductions;

iii.Issues identified in a FAR during validation to be verified during verification have not been resolved by the project proponents.

A clarification request (CL) shall be raised if information is insufficient or not clear enough to determine whether the applicable VCS requirements have been met.

The verification team identified 1 CL and no CAR in this monitoring period. The CL was satisfactorily addressed by the project proponents in the updated monitoring report version 02 dated 08/09/2022 (refer to Appendix D for further details).

2.5.1 Forward Action Requests

A forward action request (FAR) is issued for actions if the monitoring and reporting require attention and/or adjustment for the next monitoring period.

CTI confirmed that there was no FAR identified in previous verification/24/, and no FAR was raised during this verification.

2.6 Eligibility for Validation Activities

The project has been validated by TÜV Rheinland (China) Ltd. under the CDM rules and registered as CDM project on 28/09/2012, and was registered as VCS project under VCS standard version 3 through gap validation by TÜV Rheinland (China) Ltd.

CTI was accredited in Sectoral Scope 1 on validation and verification and thus is eligible for verification activities on the project. By checking the UNFCCC website, the VCS pipeline and the previous verification report, it is confirmed the verification for this monitoring period 21/12/2017-07/11/2021 which is no more than consecutive 6 years is the first verification undertaken by CTI. Therefore, the verification undertaken by CTI is in line with the requirement for VVB stipulated in Section 4.1.20 of VCS standard Version 4.3

3 VALIDATION FINDINGS

3.1 Participation under Other GHG Programs

The project has been registered as a CDM project on 28/09/2012 under the UNFCCC with Ref. 6697. The renewable CDM crediting period was selected staring on 28/09/2012. By checking the UNFCCC website: https://cdm.unfccc.int/Projects/DB/TUEV-RHEIN1342414627.77/view, it is confirmed that the project has not applied for CERs issuance under CDM.

3.2 Methodology Deviations

The validation process /16/ has assessed all factors and issues that constitute the basis for emission reductions from the project according to the applicable CDM methodology ACM0002 (version 12.2.0) /36/. There was not any methodology deviation applied to this project. Details refer to section 4.1.

3.3 Project Description Deviations

By checking the previous VCS verification reports with the monitoring period 08/11/2011-27/09/2012, the Commissioning report of the project /26/, Daily operation and maintenance records/7/ and through the site visit interview with project owner, it is confirmed that the project started operation on

08/11/2011 which is the project start date according to the requirement of VCS. A deviation was identified for the crediting period of the Project. The project is registered as VCS project under VCS version 3 and completed validation before 19/03/2020. As per VCS requirement, it remains eligible to apply the crediting period requirements under VCS Version 3 which shall be a maximum of ten years and may be renewed at most twice. Therefore, the first renewable crediting period of the Project under VCS is updated from 08/11/2011-27/09/2012 to 08/11/2011-07/11/2021, which lasts for 10 years. The project is also a registered CDM project with 3*7 renewable crediting period starting on 28/09/2012, the VCS issuance is not eligible for beyond the end of those 21 years, furthermore, considering the 20-year expected lifetime of the project, given that the VCS crediting period started on 08/11/2011, the total VCS crediting period will end on 07/11/2031.

CTI confirms that the project remains in compliance with VCS Program rules and the deviation on the crediting period has no impact on the applicability of the methodology, additionality of project activity, and the appropriateness of the baseline scenario.

3.4 Grouped Project

The project was not a grouped project; hence this clause is not applicable.

4 VERIFICATION FINDINGS

This section summarizes the findings from the verification of the emission reductions reported for the "Qingdao Longxin Wind Power Project Phase I" for the monitoring period 21/12/2017-07/11/2021.

4.1 Project Implementation Status

Project Implementation in accordance with the registered project design document

The project is a wind power grid-connected plant and locates at Liyuan Town and Dianzi Town, Pingdu City, Shandong Province of P.R.China. The geographical coordinates are verified by GPS system as 119.8827°E to 119.9503°E and 36.8157°N to 37.8915°N and the central geographical coordinates are 119.91125°E and 36.8568°N. By checking the FSR /27/, equipment purchase contract/29/ and through on-site inspection, it is confirmed that the total installed capacity is 49.8MW equipped with 24 sets of wind turbine generator in type of V90-2.0MW, with unit capacity of 2MW and 1 set of wind turbine generator in type of V90-1.8MW, with unit capacity of 1.8MW and the plant load factor is 22.70%. It was confirmed by checking the Commissioning report /26/, the Daily operation and maintenance records /7/, the previous VCS monitoring report and verification report /23/-/24/ that the project was put into operation since 08/11/2011. The actual implementation of the project during this verification period was verified in terms of nameplates of wind turbines /25/ and the equipment purchase contract /29/. The details of the turbines with respect to their installation and capacity have been verified to be consistent with description indicated in the registered VCS PD.

The electricity generated by the project activity is supplied to the North China Power Grid, which can be confirmed by the Power purchase agreement (PPA) signed between China Resources Wind Power (Qingdao) Co., Ltd. and Shandong Electric Power Company Qingdao Branch Company/3/. All the monitoring system in operation period is consistent with the description in the monitoring plan. The control system at the power plant is automated and assures continuous operation, including monitoring on malfunction of equipment. By checking the daily operation and maintenance records/7/ and via on-site visit, CTI can confirm that no serious malfunction happened and the plant was under a normal operation as expected in this monitoring period.

On-site training for the related procedures including, recording and reporting was verified to be in place/6/ and their implementation was confirmed by interview with the key operators/37/ and observing the operation.

CTI confirms that the project implementation is in accordance with the project description contained in registered VCS PD. Furthermore, through visual inspection and document review, it is confirmed that all physical features of the project activity including data collection systems and storage systems have been implemented in accordance with the monitoring plan.

Participation or any rejection under any other GHG programs, any other form of environmental credit or emissions trading program since validation or previous verification

By checking the UNFCCC website: https://cdm.unfccc.int/Projects/DB/TUEV-RHEIN1342414627.77/view, it is confirmed that the project has not applied for CERs issuance under CDM.

By interviewing with PP during on-site visit, it is confirmed that emission reductions during the period 28/09/2012-20/12/2017 is under verification by another VVB.

Therefore, it is concluded no double counting issuance has occurred for this verified monitoring period 21/12/2017-07/11/2021 under both VCS and CDM programmes. Via the on-site interview, the project proponent confirmed the emission reductions for this verified period are only requested for VCUs issuance.

The verification team has also checked other GHGs programs and forms of credit such as exchange platform of Chinese Certified Emission Reduction (CCER) projects, Gold Standard, EU ETS, IREC, China ETS and based on the interview with the project proponent, during this monitoring period, it is confirmed that except CDM and VCS scheme:

- No evidence showing that the project has participated or been rejected under any other GHG programs since validation or previous verification.
- There is no observation that the project has received or sought any other emission trading program and form of environmental credit, or has become eligible to do so since validation or previous verification.

The project is completely voluntary CDM and VCS project, not included in an emissions trading program or any other mechanism that includes GHG allowance trading. Although the national emission trading scheme has been launched since 16/07/2021, it only covers the high-emission and energy intensive industries that emit at least 26,000 tonnes of CO_2e/year as per the regulation/19/ including thermal power generation, petrochemical, chemical, building materials, iron and steel, etc. By checking the business license/14/of the project proponent, it is confirmed that the PP is qualified in business scope of renewable energy investment, thus is not included the mandatory emission control scheme and there is no emission cap enforced for the PP by further checking the compulsive company list under China ETS in public information/20/.

Moreover, it is confirmed by CTI the project is not registered as a CCER project in China, the emission reductions generated by the project activity are not eligible for transaction under China ETS as per the regulation/19/. Also, CCER mechanism has been suspended by Chinese government since 2017/20/, the project activity will not and is not allowed to registered under CCER mechanism. Thus, the assessment team concluded that the project activity not involved on other emissions trading programs and other binding limits.

The Statement on no double counting made by PP /30/ submitted by PP was checked by the verification team and confirmed to be correct in terms of the regulations and database of CCER and China ETS.

SD contributions resulted by implementation of the project activity

Via the on-site inspection, and through the interviews with the project proponent, local villagers and official from local environment bureau, and by checking the evidences provided by the project proponent, the verification team agrees on the following points regarding to the promotion of sustainability development by the project activity:

- *SDG 7, Indicator 7.2.1 Renewable energy share in the total final energy consumption*
  The project provides clean and renewable energy through making use of wind power, displacing fossil fuel power in the gird North China Power Grid. During this monitoring period, 393,219.283MWh of net electricity generated by wind energy was delivered to the North China Power Grid. It is verified by checking the daily and monthly reading records/8/ and via on-site inspection. Meanwhile, by checking the approved VCS monitoring report for the previous monitoring, it is confirmed 481,636.843MWh of cumulative renewal electricity has been supplied to the grid since the project operation. This contributes to increasing the share of clean energy consumption of China's National Plan on Implementation of the 2030 Agenda for Sustainable Development /31/.
- *SDG 8, Indicator 8.5.2 Unemployment rate by sex, age and persons with disabilities*
  It is verified by interviewing with project proponent, staff and local villagers and checking the Staff list provided by PP /21/, that the project activity provides 26 long-term job opportunities for local residents including 6 positions for female which contributes to decrease the unemployment rate of the local community. Hence, the contribution to this SDG has been confirmed and the provisions for monitoring and reporting this goal is verified as reasonable. This also contributes to "Increase labor force participation rate through implementation of the classification policy. Vigorously enforce the Law on Promotion of Employment" as stated in China's National Plan on Implementation of the 2030 Agenda for Sustainable Development /31/.
- *SDG 13, Indicator Tonnes of greenhouse gas emissions avoided or removed*

As verified by on-site inspection, and via checking the emission reduction calculation sheet/2/ and the monthly reading records/8/, 366,046 tCO_2e GHG emission reductions are generated by the project activity during this monitoring period. And by checking the approved VCS monitoring report, it is confirmed the cumulative emission reductions are 448,348tCO_2e since operation of the project activity. This contributes achieve one of China's stated sustainable development priorities "Actively adapt to climate change and strengthen resistance capacity to climate risks in agriculture, forestry, water resources and other key fields, as well as cities, coastal regions and ecologically vulnerable areas" stated in China's National Plan on Implementation of the 2030 Agenda for Sustainable Development /31/.

In conclusion, Verification Team was able to confirm that the project implementation is in accordance with the project description contained in the VCS PD. All required equipments and procedures are available and implemented in an appropriate manner. All necessary monitoring instruments are installed and operated. The project is completely operational and the same has been confirmed on-site. Neither mistakes nor malfunction on main meters have been observed during this monitoring period.

4.2 Safeguards

4.2.1 No Net Harm

Via on-site inspection and checking the EIA summary and conclusion provided in the approved VCS PD, it is confirmed that the impact caused by the project activity on the surrounding ecosystem and residents, ambient air, noise, electromagnetic impact, solid waste and wastewater etc. is insignificant, there would be no net harm caused due to the project activity. Appropriate mitigation measures have been undertaken by the project proponent to avoid the potential impacts during the construction and operation. The EIA of the project are approved by the government.

Also, no potential environment or social economic matter was found during the site visit. It is verified by interviewing with project proponent, staff and local villagers and checking the Staff list /21/, that the project has contributed to the local employment by providing job opportunities for both males and females. As stated by the plant staff during the on-site interviews, they have gained knowledge and skills in the regular training organized by the project proponent, so it can be judged that the project has, to some extent, improved the marketability of the employees as well as the possibility of lifelong learning for them.

In conclusion, the implementation of the project activity does not cause any significant negative environmental or socio-economic impacts and is beneficial to the local socio-economic environment.

4.2.2 Local Stakeholder Consultation

The local stakeholder's meeting and survey was held around the project location on 20/03/2011. Representatives from the local government and local residents attended the meeting. The project owner introduced the proposed project and the opinions expressed by the stakeholders were recorded and are available on request. Total 50 questionnaires were given out and 49 of them were returned successfully.

The stakeholder meeting and the questionnaire survey showed that the proposed project receives strong support from the local community. They all believe the proposed project will promote local economic development and agree with the project development and construction.

For continuous communication with local stakeholders, through on-site inspection and interview during on-site visit, it is confirmed that the project owner has published its phone number of the central control room at the power plant gate to local people and village committees, anyone who has concerns can call the phone number to express their opinions directly. Through interviewing with staff from local environmental protection bureau and local residents during the site visit, it is confirmed by the verification team that during the implementation stage of this monitoring period, local authority has conducted spot checks on the implementation of the project periodically as per the request from the local governments' regulations, no negative comments and issues from the local stakeholders during this monitoring period were identified and the project passed all the periodic spot checks by local government.

All such conclusion has been verified through site visit and checking VCS PD.

4.3 AFOLU-Specific Safeguards

For non-AFOLU projects, this section is not required.

4.4 Accuracy of GHG Emission Reduction and Removal Calculations

Compliance of monitoring plan with monitoring methodology

CTI is able to confirm that the registered monitoring plan is in accordance with the approved methodology applied by the project activity, i.e. ACM0002 (version 12.2.0).

Compliance of monitoring with the monitoring plan

CTI confirms that the parameters stated in the monitoring plan are monitored and reported appropriately. Parameters required to be monitored by the monitoring plan as per the monitoring methodology ACM0002 (version 12.2.0) and the management system was assessed during the site visit. The monitoring report lists each parameter required by the monitoring plan and the information flow (i.e. from data generation, recording, aggregation, calculation and reporting) for each parameter is provided. The information flow for each parameter in further verified in the following sections.

Parameters fixed ex ante

"Data and parameters fixed ex-ante" in the MR are checked against the registered CDM PDD. The parameter $EF_{grid,CM,y}$ is the combined emission factor of the grid which was determined ex-ante at the validation stage and has been fixed during the first crediting period. CTI verified and confirms that the emission factor used in the monitoring report is in compliance with the registered CDM PDD. Other ex-ante parameters (i.e. $FC_{i,m,y}$, $NCV_{i,y}$, $EF_{CO2,i,y}$, Installed Capacity, GEN_y and Efficiency of the best technology commercially) were used to calculate the parameter $EF_{grid,CM,y}$ in the registered CDM

PDD, which were validated in the validation process. CTI verified and confirms that the emission factor used in the monitoring report is in compliance with the registered CDM PDD.

Parameters monitored

CTI conducted document review and performed on-site assessment with project stakeholders to:

- A review of information flows for generating, aggregating and reporting the monitoring parameters;
- Determine whether the operational and data collection procedures are implemented in accordance with the registered monitoring plan;
- A cross-check between information provided in the monitoring report and data from other sources such as plant logbooks, sales receipts and electricity confirmation;
- An identification that quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

According to the monitoring plan, the parameters to be monitored are as below:

$EG_{output,y}$	Total electricity supplied to the Grid by the Project and Phase II project
$EG_{another,y}$	Electricity generated by the Phase II project
$EG_{import,y}$	Total electricity imported from the Grid by the Project and Phase II project through the 110 KV line
$EG_{im\text{-}spare,y}$	Electricity imported from the Grid by the project and Phase II Project through a spare 10kV line
$EG_{facility,y}$	Net electricity supplied to the Grid by the project

<u>Monitored parameter $EG_{output,y}$, $EG_{another,y}$, $EG_{import,y}$, $EG_{im\text{-}spare,y}$, $EG_{facility,y}$</u>

By site inspection and checking the registered monitoring plan and previous verification reports, CTI confirmed that the meters involved in the project are:

Meter M1 with an accuracy level of 0.5S installed at Tangtian Substation is owned by grid company to measure $EG_{output,y}$ and $EG_{import,y}$ and the sales receipts are issued by the grid company based on the readings of M1.

Meter M2 with accuracy class of 0.5S installed at the 110kV outline of the main transformer of the project on-site is owned by PP to measure the electricity supplied to the grid by the Project. The reading of M2 is carried out by the project proponent for internal quality control which is not relating to the monitoring of any parameter to be monitored required by the monitoring plan.

Meter M3 with an accuracy level of 0.5S is installed 110kV outline of the main transformer of Phase II project which is owned by the same Project Proponent as this project by checking the Business license and interviewing with PP during on-site visit. M3 is owned by PP to monitor $EG_{another,y}$.

M4 with an accuracy level of 1.0 installed the spare 10kV line is owned by PP to monitor $EG_{im\text{-}spare,y}$, and after confirmed by the grid company, sales receipts of $EG_{im\text{-}spare,y}$ are issued by the grid company based on the readings of M4.

According to the monitoring plan, net electricity supplied to the Grid by the project ($EG_{facility,y}$) is calculated as follows:

$EG_{facility,y} = EG_{output,y} - EG_{another,y} - EG_{import,y} - EG_{im\text{-}spare,y}$

The verification team had on-site checked the location of the meters against the diagram of power connection system /4/ and found them to be consistent and installed in accordance with the monitoring plan. All the monitoring facilities and system have been verified by CTI during on-site visit.

By checking Daily operation and maintenance records /7/ and via on-site visit, it is confirmed that all meters were working well during this monitoring period. Via the on-site interview and by checking the daily operation and maintenance records/7/, daily and monthly reading records/8/ electricity sales receipts /9/ and Electricity confirmation/10/ issued by grid company, it is confirmed by the verification team that $EG_{output,y}$, and $EG_{import,y}$ are continuously measured daily recorded and monthly aggregated by the gird company, the grid company supplies the reading records and the sales receipts of $EG_{output,y}$, and $EG_{import,y}$ to the project proponent every month and the sales receipts have been used for cross-checking with the monthly reading records. $EG_{another,y}$ and $EG_{im\text{-}spare,y}$ are continuously measured, daily recorded and monthly aggregated by PP. After confirmed by both sides, the sales receipts of $EG_{im\text{-}spare,y}$ were issued and are used to cross-check the monthly reading records. During this monitoring period, the cut-off time of $EG_{output,y}$, $EG_{another,y}$, $EG_{import,y}$, $EG_{im\text{-}spare,y}$ is as follows verified by checking the PPA signed between PP and the grid company as well as interview during on-site visit:

- From the start of this monitoring period to April 2020, the cut-off time is at 24:00 on the 20th day of each month;
- From May 2020 and after, the cut-off time is at 24:00 on the last day of each month.

Data in the monthly reading records were used to the report, through a cross check with sales receipts and electricity confirmation, the most conservative values were used for the monitoring report and ERs calculation spreadsheet, which has been verified by the verification team. Supporting references and data required to determine the quantity of net electricity generation supplied by the project is found to be complete and transparent.

Monitoring equipment and calibration

The documents review was carried out. Particular attention was paid to the frequency of measurements, the quality of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the registered VCS PD and the methodology ACM0002(version 12.2.0) and corresponding tool(s), and the quality assurance and quality control procedures.

All monitoring meters have been calibrated annually as per the monitoring plan covering this monitoring period. The relevant information of the calibration is listed as below:

<table>
<tr><th>Monitoring equipment</th><th>Type</th><th>Serial NO.</th><th>Accuracy</th><th>Calibration frequency</th><th>Calibration date</th><th>Validity date</th><th>Calibration entity</th></tr>
<tr><td>M1 (old)</td><td>DTZ666</td><td>3730001000000188595727</td><td>0.5S</td><td>Annually</td><td>03/04/2017
02/04/2018
01/04/2019
30/03/2020
29/03/2021</td><td>02/04/2018
01/04/2019
31/03/2020
29/03/2021
18/06/2021</td><td rowspan="5">State Grid Shandong Electric Power Metering Centre</td></tr>
<tr><td>M1 (new)</td><td>DTZ666</td><td>3730001000000580579585</td><td>0.5S</td><td>Annually</td><td>10/05/2021</td><td>09/05/2022</td></tr>
<tr><td>M2</td><td>DTSD341</td><td>11060727990139</td><td>0.5S</td><td>Annually</td><td rowspan="3">03/04/2017
02/04/2018
01/04/2019
30/03/2020
29/03/2021</td><td rowspan="3">02/04/2018
01/04/2019
31/03/2020
29/03/2021
28/03/2022</td></tr>
<tr><td>M3</td><td>DTSD341</td><td>11060727990140</td><td>0.5S</td><td>Annually</td></tr>
<tr><td>M4</td><td>DTZ522</td><td>370010042337325</td><td>1.0</td><td>Annually</td></tr>
</table>

The originally used meter M1 (SN. QD0243006231) was replaced by M1 (old) with SN. 3730001000000188595727 on 18/10/2016 with the consent by the grid company before this monitoring period, and the accuracy of M1 (old) with SN.3730001000000188595727 was confirmed to be 0.5S which is the same as the originally used M1 and also in line with the requirement of monitoring plan by checking the meter calibration reports and interview with PP during on-site visit.

The originally used meter M4 (SN. DT4803509) was replaced by currently used M4 (SN.370010042337325) on 11/10/2015 with the consent by the grid company before this monitoring period, CTI confirmed through on-site interview with PP and checking the meter calibration reports that the accuracy of the currently used M4 is 1.0 which is the same as the originally used M4 and in line with the accuracy requirement of the monitoring plan.

During this monitoring period, the old meter M1 (SN. 3730001000000188595727) was replaced with the new M1 (SN. 3730001000000580579585) on 18/06/2021. CTI confirmed through on-site interviews with PP and by reviewing the qualification documents of State Grid Shandong Electric Power Metering Centre/12/and the meter calibration reports/13/ that the meter replacement was carried out with the consent by the grid company and the new meter’s accuracy is 0.5S which is the same as the old meter and thus, still met the accuracy requirements of the monitoring plan.

Calibration records and accreditation certificates/12//13/have been verified by the verification team. The actual accuracy for M1(old and new), M2 and M3 is verified to be 0.5S which is in line with the requirement in monitoring plan. The actual accuracy for meter M4 is verified to be 1.0 which is higher than the requirement in the monitoring plan and the accuracy of all meters is in compliance with the Technical administrative code of electric energy metering /15/. The calibration frequency of all meters is annual, which is consistent with the monitoring plan. Based on the site visit and by checking the calibration reports, CTI confirms that:

- All meters have been installed in accordance with the monitoring plan;
- The meters' accuracy is in line with the requirement of the monitoring plan;
- The calibration frequency of the meters is annual, which is in line with the monitoring plan and national technical standard/15/ as required by VCS standard. And the calibrations are verified to be valid for the whole reporting period.

CL01 was raised and successfully closed, please refer to Appendix D for details.

Data management and control

China Resources Wind Power (Qingdao) Co., Ltd. is responsible for operation and routine maintenance of power plant under the project activity. The quality assurance and quality control procedures have been addressed in the project management and monitoring manual/5/, including the organization structure with the responsibilities, personnel competencies, monitoring procedures and monitoring management. By interview with the staff/37/ and check records /7//8/ during on-site visit, it can be confirmed that the monitoring management system is implemented following the project management and monitoring manual.

All monitoring devices have been calibrated and maintained periodically to ensure the accuracy of measurement. Calibration records of instruments used in measurements were made available during the verification visit and found to be valid for the entire period of the verification. Competence and training records of in-plant personnel engaged in measurement of plant parameters were presented during verification and found to be in order.

As the above and document review, verification team considers that the project activity operated properly in the way complying with the implementation plan and monitoring plan and the monitoring was performed in a transparent and comprehensive manner during this monitoring period.

Emission reduction calculation

CTI confirms that appropriate methods and formulae for calculating baseline emissions, project emissions and leakage have been followed, and the assumptions, emission factors and default values that are applied in the calculation have been justified.

According to the applied methodology, the emission reductions are determined as the difference between the baseline emissions, project emissions and leakage:

$ER_y = BE_y - PE_y - L_y$

Baseline emissions

Baseline emissions are determined as multiplying quantity of net electricity supplied to the Grid by the project ($EG_{facility,y}$) by the validated ex-ante fixed grid emission factor ($EF_{grid,CM,y}$).

$BE_y = EG_{facility,y} \times EF_{grid,CM,y}$

Combined margin CO_2 emission factor ($EF_{grid,CM,y}$)

$EF_{grid,CM,y}$ is the grid emission factor of the which has been verified ex-ante in the validation stage in the VCS PD as 0.9309 tCO_2e/MWh.

Net electricity supplied to the Grid by the project ($EG_{facility,y}$)

As per the Section 4.1 above, EG_y is calculated as:

$EG_{facility,y} = EG_{output,y} - EG_{another,y} - EG_{import,y} - EG_{im\text{-}spare,y}$

The verification team has checked the daily and monthly reading records, sales receipts and electricity confirmation issued by the grid company and emission reduction calculation sheet, and confirmed the data and calculation applied are correct, complete and transparent. The monthly electricity data and $EG_{facility,y}$ calculation are summarized in the Tables as follow:

Table 1 Monitored monthly values of $EG_{output,y}$, $EG_{another,y}$, $EG_{import,y}$, $EG_{im\text{-}spare,y}$ and calculation of $EG_{facility,y}$

From	To	Total electricity supplied to the Grid by the Project and Phase II Project ($EG_{output,y}$)			Total electricity imported from the Grid by the project and Phase II project through the 110kV line ($EG_{import,y}$)			Electricity imported from the Grid by the project and Phase II project through a spare 10 kV line （$EG_{im,spare\ y}$）			Electricity generated by the Phase II Project ($EG_{another,y}$)	Net electricity supplied to the grid by the project ($EG_{facility}$)
		unit: MWh			unit: MWh			unit: MWh			unit: MWh	unit: MWh
		Values from meter readings of M1	Values from sales receipts	Conservative values	Values from meter readings of M1	Values from sales receipts	Conservative values	Values from meter readings of M4	Values from sales receipts	Conservative values	Values from meter readings of M3	Calculated as $EG_{output,y}$- $EG_{import,y}$- $EG_{im\text{-}spare,y}$- $EG_{another,y}$
21/12/2017	31/12/2017*	7,835.520	7,835.520	7,835.520	13.068	13.068	13.068	0.169	0.169	0.169	3,150.400	4,671.883
Subtotal 2017				**7,835.520**			**13.068**			**0.169**	**3,150.400**	**4,671.883**
01/01/2018	20/01/2018*	11733.480	11733.480	11,733.480	22.572	22.572	22.572	0.292	0.292	0.292	6,223.360	5,487.256
21/01/2018	20/02/2018	19946.520	19946.520	19,946.520	54.120	54.120	54.120	0.461	0.461	0.461	10,283.680	9,608.259
21/02/2018	20/03/2018	25215.960	25215.960	25,215.960	46.200	46.200	46.200	0.461	0.461	0.461	13,840.640	11,328.659
21/03/2018	20/04/2018	26242.920	26242.920	26,242.920	19.800	19.800	19.800	0.461	0.461	0.461	13,553.760	12,668.899
21/04/2018	20/05/2018	23374.560	23374.560	23,374.560	23.760	23.760	23.760	0.461	0.461	0.461	11,740.960	11,609.379
21/05/2018	20/06/2018	17784.360	17784.360	17,784.360	43.560	43.560	43.560	0.461	0.461	0.461	10,234.400	7,505.939
21/06/2018	20/07/2018	19788.120	19788.120	19,788.120	39.600	39.600	39.600	2.877	2.877	2.877	10,816.960	8,928.683
21/07/2018	20/08/2018	18961.800	18961.800	18,961.800	80.520	80.520	80.520	0.461	0.461	0.461	9,928.160	8,952.659
21/08/2018	20/09/2018	11412.720	11412.720	11,412.720	62.040	62.040	62.040	0.461	0.461	0.461	5,241.280	6,108.939
21/09/2018	20/10/2018	12418.560	12418.560	12,418.560	96.360	96.360	96.360	0.461	0.461	0.461	5,292.320	7,029.419
21/10/2018	20/11/2018	13625.040	13625.040	13,625.040	93.720	93.720	93.720	0.461	0.461	0.461	12,615.680	915.179
21/11/2018	20/12/2018	13000.680	13000.680	13,000.680	50.160	50.160	50.160	0.461	0.461	0.461	4,565.440	8,384.619
21/12/2018	31/12/2018*	11433.840	11433.840	11,433.840	30.492	30.492	30.492	0.169	0.169	0.169	4,158.880	7,244.299
Subtotal 2018				**224,938.560**			**662.904**			**7.948**	**118,495.520**	**105,772.188**
01/01/2019	20/01/2019*	4994.880	4994.880	4994.880	52.668	52.668	52.668	0.292	0.292	0.292	823.680	4,118.240
21/01/2019	20/02/2019	18499.800	18499.800	18499.800	35.640	35.640	35.640	0.461	0.461	0.461	10917.280	7,546.419
21/02/2019	20/03/2019	15626.160	15626.160	15626.160	73.920	73.920	73.920	0.461	0.461	0.461	8004.480	7,547.299
21/03/2019	20/04/2019	26508.240	26508.240	26508.240	25.080	25.080	25.080	0.461	0.461	0.461	13712.160	12,770.539
21/04/2019	20/05/2019	19852.800	19852.800	19852.800	22.440	22.440	22.440	0.461	0.461	0.461	8544.800	11,285.099

21/05/2019	20/06/2019	24132.240	24132.240	24132.240	14.520	14.520	14.520	0.461	0.461	0.461	14569.280	9,547.979
21/06/2019	20/07/2019	13248.840	13248.840	13248.840	52.800	52.800	52.800	0.461	0.461	0.461	5536.960	7,658.619
21/07/2019	20/08/2019	12788.160	12788.160	12788.160	56.760	56.760	56.760	0.461	0.461	0.461	7145.600	5,585.339
21/08/2019	20/09/2019	7326.000	7326.000	7326.000	76.560	76.560	76.560	2.229	2.229	2.229	5456.000	1,791.211
21/09/2019	20/10/2019	11754.600	11754.600	11754.600	66.000	66.000	66.000	0.461	0.461	0.461	6908.000	4,780.139
21/10/2019	20/11/2019	19633.680	19633.680	19633.680	96.360	96.360	96.360	0.461	0.461	0.461	8101.280	11,435.579
21/11/2019	20/12/2019	18122.280	18122.280	18122.280	50.160	50.160	50.160	0.461	0.461	0.461	5660.160	12,411.499
21/12/2019	31/12/2019*	6485.160	6485.160	6485.160	18.392	18.392	18.392	0.169	0.169	0.169	5079.360	1,387.239
Subtotal 2019				198,972.840			641.300			7.300	100,459.040	97,865.200
01/01/2020	20/01/2020*	7656.000	7656.000	7656.000	31.768	31.768	31.768	0.292	0.292	0.292	6156.480	1,467.460
21/01/2020	20/02/2020	14226.960	14226.960	14226.960	64.680	64.680	64.680	0.461	0.461	0.461	7705.280	6,456.539
21/02/2020	20/03/2020	21951.600	21951.600	21951.600	47.520	47.520	47.520	0.461	0.461	0.461	10456.160	11,447.459
21/03/2020	20/04/2020	20577.480	20577.480	20577.480	43.560	43.560	43.560	0.461	0.461	0.461	11667.040	8,866.419
21/04/2020	31/05/2020	37480.080	37480.080	37480.080	52.800	52.800	52.800	0.461	0.461	0.461	19534.240	17,892.579
01/06/2020	30/06/2020	14898.840	14898.840	14898.840	68.640	68.640	68.640	0.461	0.461	0.461	7868.960	6,960.779
01/07/2020	31/07/2020	13419.120	13419.120	13419.120	51.480	51.480	51.480	0.461	0.461	0.461	6988.960	6,378.219
01/08/2020	31/08/2020	13596.000	13596.000	13596.000	73.920	73.920	73.920	1.834	1.834	1.834	7129.760	6,390.486
01/09/2020	30/09/2020	9273.000	9273.000	9273.000	71.280	71.280	71.280	0.461	0.461	0.461	4847.040	4,354.219
01/10/2020	31/10/2020	9403.680	9403.680	9403.680	112.200	112.200	112.200	0.461	0.461	0.461	4875.200	4,415.819
01/11/2020	30/11/2020	17472.840	17472.840	17472.840	42.240	42.240	42.240	0.461	0.461	0.461	9056.960	8,373.179
01/12/2020	31/12/2020	18115.680	18115.680	18115.680	46.200	46.200	46.200	0.461	0.461	0.461	9278.720	8,790.299
Subtotal 2020				198,071.280			706.288			6.736	105,564.800	91,793.456
01/01/2021	31/01/2021	24944.040	24944.040	24944.040	44.880	44.880	44.880	0.461	0.461	0.461	13113.760	11,784.939
01/02/2021	28/02/2021	18798.120	18798.120	18798.120	25.080	25.080	25.080	0.461	0.461	0.461	9697.600	9,074.979
01/03/2021	31/03/2021	19070.040	19070.040	19070.040	43.560	43.560	43.560	0.461	0.461	0.461	10167.520	8,858.499
01/04/2021	30/04/2021	20931.240	20931.240	20931.240	22.440	22.440	22.440	0.461	0.461	0.461	11142.560	9,765.779
01/05/2021	31/05/2021	27313.440	27313.440	27313.440	21.120	21.120	21.120	0.461	0.461	0.461	13831.840	13,460.019
01/06/2021	30/06/2021	19107.000	19107.000	19107.000	31.680	31.680	31.680	0.461	0.461	0.461	10343.520	8,731.339

01/07/2021	31/07/2021	19459.440	19459.440	19459.440	22.440	22.440	22.440	0.461	0.461	0.461	10243.200	9,193.339
01/08/2021	31/08/2021	9984.480	9984.480	9984.480	95.040	95.040	95.040	0.461	0.461	0.461	5220.160	4,668.819
01/09/2021	30/09/2021	16733.640	16733.640	16733.640	44.880	44.880	44.880	2.722	2.722	2.722	8523.680	8,162.358
01/10/2021	31/10/2021	12858.120	12858.120	12858.120	79.200	79.200	79.200	0.461	0.461	0.461	6633.440	6,145.019
01/11/2021	07/11/2021*	5511.968	5511.968	5511.968	33.000	33.000	33.000	0.461	0.461	0.461	2207.040	3,271.467
Subtotal 2021				194,711.528			463.320			7.332	101,124.320	93,116.556
Total				824,529.728			2,486.880			29.485	428,794.080	393,219.283

*Note:

1. Monthly readings of $EG_{output,y}$, $EG_{import,y}$ and $EG_{im\text{-}spare,y}$ are cross-checked with the sales receipts issued by grid company, the most conservative values are applied for calculation of $EG_{facility,y}$.

2. Readings of $EG_{another,y}$ is checked by CTI with the original copy of the reading records and confirmed that the values used for ER calculation is correct and complete.

3. All meters' readings for the periods: 21st -31st Dec of 2017 to 2019, 01st -20th Jan of 2018 to 2019 and 01st -07th Nov of 2021 are sourced from the daily reading records and the readings of $EG_{output,y}$, $EG_{import,y}$ and $EG_{im\text{-}spare,y}$ have been verified with the confirmation /10/ issued by the grid company.

The corresponding baseline emissions during this monitoring period is calculated as:

Year	$EG_{facility,y}$ (MWh)	$EF_{grid,CM,y}$ (tCO_2e/MWh)	BE_y (tCO_2e)
2017 (21/12/2017-31/12/2017)	4,671.883	0.9309	4,349
2018	105,772.188	0.9309	98,463
2019	97,865.200	0.9309	91,102
2020	91,793.456	0.9309	85,450
2021 (01/01/2021-07/11/2021)	93,116.556	0.9309	86,682
Total	**393,219.283**	-	**366,046**

Project emissions

As statement in the registered VCS PD, for the wind power activities, the project emissions from the project is zero. Hence, PE_y during the monitoring period from Qingdao Longxin Wind Power Project Phase I is zero.

Leakages

Leakage does not need to be accounted for this project as per the monitoring methodology ACM0002 (version 12.2.0).

Emission reductions

The emission reductions for this monitoring period were calculated as:

Monitoring period	GHG emission reductions or removals (tCO_2e)
21/12/2017 – 31/12/2017	4,349
01/01/2018 – 31/12/2018	98,463
01/01/2019 – 31/12/2019	91,102
01/01/2020 – 31/12/2020	85,450
01/01/2021 – 07/11/2021	86,682
Total ERs claimed	**366,046**

4.5 Quality of Evidence to Determine GHG Emission Reductions and Removals

The assessment of each parameter related to calculate the GHG emission reductions is justified in Sections 4.1 and 4.4 above. Via the on-site inspection and desk review, the verification team verified the supporting evidence provided by the project proponent as listed in Appendix 3 used to determine the GHG emission reductions

For each reported data, the evidence is provided and verified as sufficient and quality is appropriate. Also, the cross-checks have been performed on the reported data with different source of evidence. The information flow from data generation and aggregation, to recording, calculation and final transposition into the monitoring report has been assessed by VVB in Section 4.4. Moreover, the calibration is verified as stated in Section 4.1 via on-site inspection with evidences to confirm the meters' location, accuracy and calibration frequency are in line with requirement of the monitoring plan.

Therefore, it is concluded that the evidence provided are verified as sufficient and quality is appropriate and thus the evidence can be used to determine the GHG emission reductions and removals for this monitoring period.

4.6 Non-Permanence Risk Analysis

The project is not AFOLU project, and thus non-permanence risk analysis is not applicable for the project.

5 VERIFICATION CONCLUSION

Shenzhen CTI International Certification Co., Ltd (CTI) has performed the verification of the emission reductions that have been reported for the project activity “Qingdao Longxin Wind Power Project Phase I” in China (VCS Project ID: 969) for the period 21/12/2017-07/11/2021.

The verification is based on the baseline and monitoring methodology ACM0002 (version 12.2.0), the VCS PD. The verification consisted of the following three phases: i) desk review of the project design and the baseline and monitoring plan; ii) follow-up interviews with project stakeholders; iii) resolution of outstanding issues and the issuance of the final verification and certification report.

The project proponents are responsible for the collection, calculation and determination of the GHG data in accordance with the monitoring plan and the reporting of GHG emission reductions on the basis set out within the project monitoring report.

Our verification and validation approach was based on the requirements as defined under the applicable VCS Standard Version 4.3 and relevant UNFCCC requirements. Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. The verification can confirm that:

- the project is implemented and operated as per the registered VCS PD ;
- the monitoring plan is as per the applied methodology;
- the monitoring complies with the monitoring plan;
- the monitoring report and other supporting documents provided are complete and verifiable and in accordance with the applicable VCS Standard Version 4.3 and CDM requirements;
- the installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately;
- the monitoring system is in place and generates GHG emission reductions data;
- the GHG emission reductions are calculated without material misstatements.
- the project deviation is appropriately described and justified; the deviation was verified by the verification team as valid, and the project remains in compliance with the VCS rules.

It is CTI’s responsibility to provide an independent verification statement on the reported GHG emission reductions for the project. Based on an understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these, CTI planned and performed our work to obtain the information and explanations that we considered necessary to provide reasonable assurance that reported GHG emission reductions are fairly stated.

CTI does not assume any responsibility towards the issuance and utilization of the VCUs hereby verified and certified. Request for issuance of VCUs shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to CTI and the engagement conditions detailed in this report. CTI cannot be held liable by any party for decisions made or not made based on this report.

In CTI's opinion the GHG emissions reductions of the "Qingdao Longxin Wind Power Project Phase I" for the period 21/12/2017-07/11/2021 are fairly stated in the monitoring report (version 02 dated 08/09/2022). The GHG emission reductions were calculated correctly on the basis of the approved methodology ACM0002 (version 12.2.0) and the monitoring plan contained in the registered VCS PD.

CTI can confirm that the GHG emission reductions are calculated without material misstatements. Based on the evidence and information that are considered necessary to guarantee that GHG emission reductions are appropriately calculated, CTI confirms the following statement:

Reporting period: 21/12/2017-07/11/2021

Verified GHG emission reductions and removals in the above verification period:

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
2017 (21/12/2017-31/12/2017)	4,349	0	0	4,349
2018	98,463	0	0	98,463
2019	91,102	0	0	91,102
2020	85,450	0	0	85,450
2021 (01/01/2021-07/11/2021)	86,682	0	0	86,682
Total	**366,046**	**0**	**0**	**366,046**

Feng Tian

Mr. Feng Tian

Team Leader

18-September-2022

Shunrong Lin

Ms. Lin Shunrong

Technical Reviewer

18-September-2022

APPENDIX A: ABBREVIATIONS

CAR	Corrective Action Request
CCER	Chinese Certified Emission Reduction
CER	Certified Emission Reduction(s)
CL	Clarification request
CO_2	Carbon Dioxide
CO_2e	Carbon Dioxide Equivalent
CTI	Shenzhen CTI International Certification Co., Ltd
DOE	Designated Operational Entity
EF	Emission Factor
EIA	Environmental Impact Assessment
ER	Emission Reduction
ETN	Electricity Transaction Note
EU ETS	EU Emissions Trading Scheme
FAR	Forward Action Request
FSR	Feasibility Study Report
GHG	Greenhouse Gas(es)
IREC	International Renewable Energy Certificates
MR	Monitoring Report
NEPG	Northeast Power Grid
PD	Project Description
PP	Project Proponent
SN	Serial Number
VCS	Verified Carbon Standard
VCU	Verified Carbon Unit

APPENDIX B: REFERENCES

Documentation used to verify the information provided by the project proponents

/1/	Juno Carbon Investment & Environmental Technology (Beijing) Co., Ltd.: VER/VCU Monitoring Report for Qingdao Longxin Wind Power Project Phase I, version 01 dated 01/08/2022, version 02 dated 08/09/2022
/2/	Juno Carbon Investment & Environmental Technology (Beijing) Co., Ltd.: Emission reduction calculation spreadsheet for Qingdao Longxin Wind Power Project Phase I, version 02 dated 08/09/2022
/3/	China Resources Wind Power (Qingdao) Co., Ltd. and Shandong Electric Power Company Qingdao Branch Company: Power Purchase Agreement for Qingdao Longxin Wind Power Project Phase I
/4/	China Resources Wind Power (Qingdao) Co., Ltd.: Diagram of power connection system
/5/	China Resources Wind Power (Qingdao) Co., Ltd.: VER monitoring manual and management procedure.
/6/	China Resources Wind Power (Qingdao) Co., Ltd.: Records of training for on-site staff.
/7/	China Resources Wind Power (Qingdao) Co., Ltd.: Daily operation and maintenance records.
/8/	China Resources Wind Power (Qingdao) Co., Ltd.: Daily and monthly reading records of Qingdao Longxin Wind Power Project Phase I from 21/12/2017-07/11/2021
/9/	Shandong Electric Power Company Qingdao Branch Company.: Electricity Sales Receipts from 21/12/2017-07/11/2021
/10/	Shandong Electric Power Company Qingdao Branch Company: Electricity confirmation
/11/	China Resources Wind Power (Qingdao) Co., Ltd.: Registered VSC PD (version 02 dated 30/01/2013)
/12/	China National Accreditation Service for Conformity Assessment: Accreditation certificates of State Grid Shandong Electric Power Metering Centre which validity covering this monitoring period 21/12/2017-07/11/2021
/13/	State Grid Shandong Electric Power Metering Centre: Calibration certificates for meters: M1 (old and new), M2, M3 and M4 covering this monitoring period
/14/	Business license of China Resources Wind Power (Qingdao) Co., Ltd.

/15/	State Economic and Trade Commission: Technical administrative code of electric energy metering (DL/T 448-2016)
/16/	UNFCCC, http://cdm.unfccc.int
/17/	Gold Standard, http://www.goldstandard.org/
/18/	China CCER Exchange Info-platform, http://cdm.ccchina.org.cn/ccer.aspx
/19/	Measures for the Administration of Carbon Emission Trading, issued by Ministry of Ecology and Environment of China, dated 31/12/2020
/20/	Enforced company list in China cap & trade scheme, issued by Ministry of Ecology and Environment of China
/21/	China Resources Wind Power (Qingdao) Co., Ltd.: Staff list of the project
/22/	CDM-PDD, version 03 dated 18/04/2012 TÜV Rheinland (China) Ltd.: CDM Validation Report, version 01.2, dated 16/07/2012
/23/	VER/VCU Monitoring Report for Qingdao Longxin Wind Power Project Phase I of previous monitoring period 08/11/2011-27/09/2012
/24/	TÜV Rheinland (China) Ltd.: VCS Verification Report for Qingdao Longxin Wind Power Project Phase I of previous monitoring period 08/11/2011-27/09/2012
/25/	Vestas Wind Systems A/S: Nameplates of the equipment
/26/	China Resources Wind Power (Qingdao) Co., Ltd.: Commissioning report
/27/	Shandong Provincial Industry Design Institute Co., Ltd: Feasibility Study Report of Qingdao Longxin Wind Power Project Phase I
/28/	Qingdao City Environmental Protection Science Research Institute: EIA Report of Qingdao Longxin Wind Power Project Phase I dated 09/2010 Qingdao City Environmental Protection Bureau: EIA approval issued on 12/10/2010
/29/	Co-signed by the Vestas Wind Systems A/S, and China Resources Wind Power (Qingdao) Co., Ltd.: Wind Turbines Purchase Agreement
/30/	China Resources Wind Power (Qingdao) Co., Ltd.: Statement on no double counting
/31/	China's National Plan on Implementation of the 2030 Agenda for Sustainable Development: http://www.gov.cn/xinwen/2016-10/13/5118514/files/44cb945589874551a85d49841b568f18.pdf

Methodologies, tools and other guidance

/32/	Verified Carbon Standard: VCS Standard, Version 4.3
/33/	Verified Carbon Standard: VCS Program Guide, version 4.2
/34/	Verified Carbon Standard: VCS Sectoral Scopes http://v-c-s.org/node/448
/35/	Verified Carbon Standard: Registration and Issuance Process, version 4.2
/36/	UNFCCC EB: Approved methodology, ACM0002 , version 12.2.0

Persons interviewed

	Interviewed personnel	Role	Organization
/37/	Mr. Qiu Zhenzhen	Plant Director	China Resources Wind Power (Qingdao) Co., Ltd.
	Mr. Li Jiazhuo	O&M Staff	
	Mr. Zhang Shufeng	O&M Staff	
	Mr. Man Wenyan	Local resident	Local Village
	Ms. Tanliyun		
	Mr. Yang Shaokun		
	Mr. Hou Kun		
	Mr. Li Peiyu	Staff	Local Environment Protection Bureau
	Ms. Ding Xue	Project Manager	Juno Carbon Investment & Environmental Technology (Beijing) Co., Ltd.

APPENDIX C: COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWERS

Mr. Tian FENG

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification					
GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
√	-	√	√	-	-

Scope	**Technical Area**
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.2: Energy generation from renewable energy sources

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by: Wu Lin

Wu LIN

Technical Competent Manager

Shenzhen, 10/02/2022

Ms. Shunrong LIN

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification						
Status	GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
Date	√	√	√	√	√	√

Scope	**Technical Area**
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.2: Energy generation from renewable energy sources
SS 14: Afforestation and reforestation	TA 14.1: Afforestation and reforestation
SS 15: Agriculture	TA 15.1: Agriculture

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by: Wu Lin

Wu LIN

Technical Competent Manager

Shenzhen, 01/01/2021

APPENDIX D: CORRECTIVE ACTION REQUESTS, CLARIFICATION REQUESTS AND FORWARD ACTION REQUESTS

Table 1: Corrective Action Requests

CAR ID	Corrective Action Request	Response by Project Proponent	Verification Team Assessment
NA	NA	NA	NA

Table 2: Clarification Requests

CL ID	Clarification Request	Response by Project Proponent	Verification Team Assessment
CL 01	MR (version 01 dated 01/08/2022) lacks the information on calibration date and validity of the electricity meters, thus CL01 was raised to clarify.	The information on calibration of the meters had been included in the updated in the MR.	OK. It has been revised and verified by CTI by checking updated MR and the calibration reports. Therefore, CL 01 is closed.

Table 3: Forward Action Requests

FAR ID	Forward Action Request	Response by Project Proponent	Verification Team Assessment
NA	NA	NA	NA